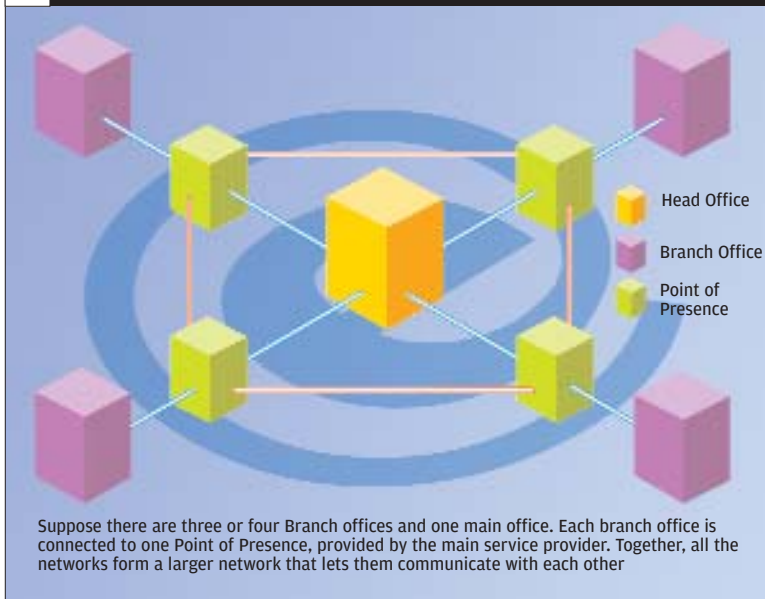


How Does A VPN Work?



Infograph: Solomon Lewis



“One shouldn’t stick to just one provider. If my leased line fails, or if the service provider is down, I can still reach by head office over any other ISP provided bandwidth, and pay only till the time I use it

Abbas Dean
IT Executive
Anglo-Eastern Ship
Management Ltd

plier of Indian Sea farers on Foreign Vessels. Set up 30 years ago, the company offers services such as ship management, technical services and crew management. The Mumbai office, with a staff of 100, takes care of a major chunk of the recruitment process.

The company is headquartered in Hong Kong. It has a long list of offices spread all over the globe—the US, Canada, New Zealand, Belgium, the UK, Japan, Philippines and Singapore. The communication network, thus, needs to be impeccable to allow the head office to keep track of all its regional branches.

The First Steps

Back in 2001, the company began reaping the benefits of technology by streamlining much of its work. During that year, Anglo-Eastern went online, offering full details of ships, including technical and financial reports. It installed an online vessel-condition and ship-management system. With such a setup, superintendents and owners could access technical reports, audits, accounting reports, and financial analyses—all over the Internet.

The next year, Anglo Eastern set up a frame relay network, connecting the head office in Hong Kong with all the regional offices. The idea was find an alternative for spending heavily on communication, without compromising on security.

A frame relay network establishes a WAN between the connected sites. The connection uses private lines or circuit-switches over leased lines. Despite a much faster rate of data transfer, there were a few problems.

According to Abbas Dean, IT Executive, Anglo-Eastern, “Whenever the connectivity to the Point of Presence (POP) through an MTNL connection snaps, as it happened to us during the monsoons, the connection simply goes down. We were helpless and without any means to communicate.” Abbas recalls those days

when additional backup lines had to be maintained. One additional line meant a constant, heavy expenditure.

For an organisation that depends heavily on round-the-clock, instant communication, this posed a serious problem. So the hunt began for another form of network—a substitute that was more secure and cost-effective.

Enter Virtual Private Network

A Virtual Private Network uses ‘virtual’ connections routed through the Internet. Unlike a typical leased line, this connection is between the company’s private network and a remote site or an employee. Imagine you are in Country A and wish to travel to Country B. The journey could be by road.

But that makes you visible to all and sundry who take the same route. Suppose, now, that Country A decides to build a bridge to the other country, keeping in mind exclusive direct access and better security. But building bridges is an expensive affair.

Given that VPNs play a similar role—that of an ‘exclusive bridge’—the next big question is, how does it tackle the expenditure angle of the whole setup? We could give the inhabitants of Country A a small plane. A plane is ideal when it comes to speedy access as well as security and privacy.

This is exactly how VPNs work. Each remote member of your network can communicate in a secure and reliable manner using the Internet as the medium (the sky or airspace, in our analogy) to connect to the private LAN (using a plane, in our example). And this is precisely what Anglo-Eastern Ship Management Ltd.

The headquarters and all the regional offices were connected by a single network. All that the Mumbai office had to do was to invest in a dedicated leased line for a short distance, right up to the service provider. The result: with just a 2 Mbps line, the Mumbai office is constantly connected to the head office and regional offices all over the world.



VPN: What To Look Out For

There are many vendors providing such a service, from VSNL to Sify. According to Abbas Dean of Anglo-Eastern, keep the following things in mind when opting for VPN:

- ❑ **Data Security:** It is always not required to spend heavily on security. Spend on security as much as the business requires. If possible, outsource security related issues to the service provider
- ❑ **Geographical Location:** It is better to select a service provider who has POP in all countries or cities of your establishment
- ❑ **Reliability:** Depends on the service provider’s infrastructure
- ❑ **Network Management:** What kind of add-ons does a service provider offer? Does he offer to manage Net traffic as well?
- ❑ **Telecom Infrastructure:** A VPN requires a good telecom infrastructure